Arista Access Node IND-ACC-76 Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) Resolution SOP

Node: IND-ACC-76 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9864

Description

This Standard Operating Procedure (SOP) is applicable to the Arista Access Node IND-ACC-76 located at the Indiranagar Exchange, which is experiencing a major optical transport failure. This failure domain is primarily caused by the degradation of SFP/XFP transceivers or fiber cuts, leading to a significant reduction in receive (Rx) and/or transmit (Tx) power levels. The underlying protocol or hardware mechanics responsible for this failure are the Optical Transport Network (OTN) protocols, which utilize SFP/XFP transceivers to establish connections between the Access Node and the Optical Transport Network (OTN) equipment. The OTN protocols, such as ODUk, OCh, and OMS, rely on the accurate transmission and reception of optical signals to ensure reliable data transport. In the event of SFP/XFP degradation or fiber cuts, the OTN protocols fail to operate correctly, leading to a major optical transport failure.

Pre-requisite Checks

- prerequisite: Confirm that the Indiranagar Exchange is physically accessible and that the remote hands have been dispatched to the site.

- prerequisite: Verify that the redundant link status is operational and can take over in case of a failure.

Potential Root Causes

Cause	Probability
SFP/XFP Transceiver Degradation	High
Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces, Expected Output: Interface status: up/down, Description: This command is used to verify the status of the interfaces on the Arista Access Node. The expected output indicates whether the interface is up or down.

Command: show interfaces detail, Expected Output: Interface details: SFP/XFP type, optical signal levels, Description: This command is used to verify the details of the interfaces on the Arista Access Node. The expected output indicates the type of SFP/XFP used and the optical signal levels.
```

```
Command: show optics, Expected Output: Optics details: SFP/XFP type, optical signal levels, Description: This command is used to verify the details of the optics on the Arista Access Node. The expected output indicates the type of SFP/XFP used and the optical signal levels.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP Transceiver, Command: replace interfaces slot with , Description: This step involves replacing the degraded SFP/XFP transceiver with a new one. The command is used to replace the SFP/XFP transceiver in the specified slot.

Step 2: Step: Verify the Fiber Connection, Command: show interfaces, Description: This step involves verifying the fiber connection to ensure that it is secure and not damaged.

Step 3: Step: Adjust the Rx/Tx Power Levels, Command: set optics rx-tx-power , Description: This step involves adjusting the receive and transmit power levels to ensure that they are within the acceptable range.

Post-Check Verification

```
Command: show interfaces, Expected Output: Interface status: up/up, Description: This command is used to verify that the interfaces are up and operational.

Command: show interfaces detail, Expected Output: Interface details: SFP/XFP type, optical signal levels, Description: This command is used to verify that the details of the interfaces are correct.

Command: show optics, Expected Output: Optics details: SFP/XFP type, optical signal levels, Description: This command is used to verify that the details of the optics are correct.
```

Roll-back Procedures

- Step: Restore the Original SFP/XFP Transceiver, Command: replace interfaces slot with , Description: This step involves restoring the original SFP/XFP transceiver that was replaced.

- Step: Re-Adjust the Rx/Tx Power Levels, Command: set optics rx-tx-power , Description: This step involves re-adjusting the receive and transmit power levels to their original settings.